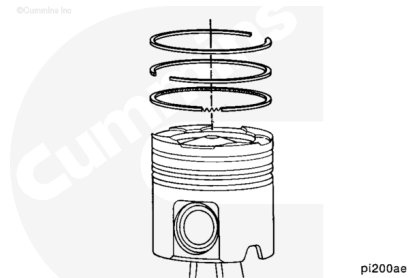


Trainings ▾

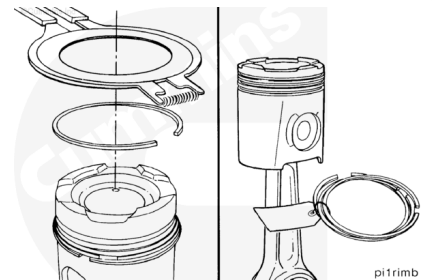
Remove

Articulated Piston

Remove the piston rings.



Place a tag on the rings, and record the cylinder number of the piston on the tag for future reference, if required.

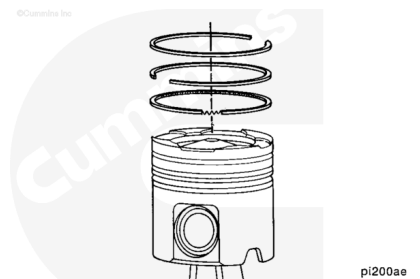


Single-Piece Piston

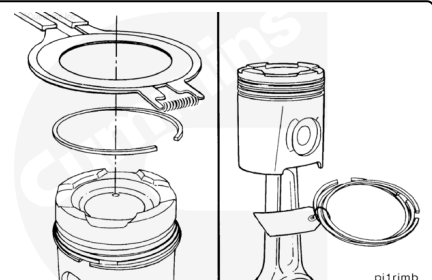
⚠ CAUTION ⚠

Do not over-expand the second piston ring on single-piece pistons.

Use a universal ring expander to remove the second piston ring on single-piece pistons.



Place a tag on the rings, and record the cylinder number of the piston on the tag for future reference, if required.

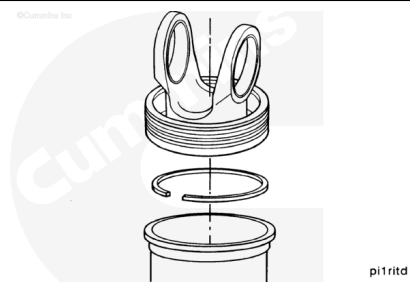


Measure

Articulated Piston

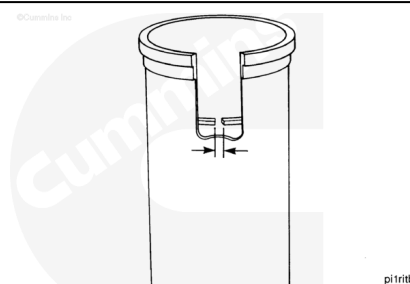
To check the ring gap, install the piston rings into the cylinder liner in which they were used below the ring reversal area.

Use the top part of the piston to correctly position the ring in the liner.



Use a feeler gauge to inspect the ring end gap.

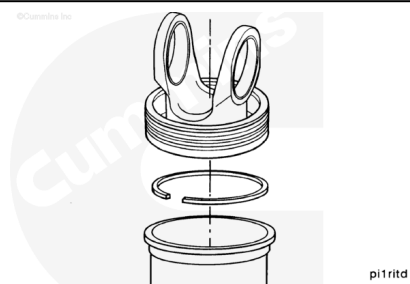
Articulated Piston Ring Gap			
	mm		in
Top	0.33	MIN	0.013
	0.53	MAX	0.021
Second	1.87	MIN	0.074
	2.13	MAX	0.084
Oil	0.21	MIN	0.008
	0.51	MAX	0.020



Single-Piece Piston

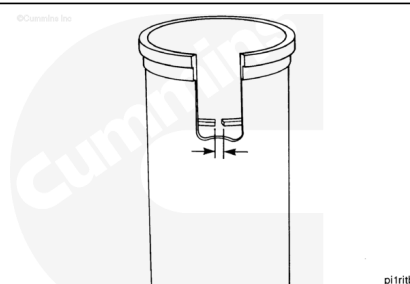
To check the ring gap, install the piston rings into the cylinder liner in which they were used below the ring reversal area.

Use the top part of the piston to correctly position the ring in the liner.



Use a feeler gauge to inspect the ring end gap.

Single-Piece Piston Ring Gap			
	mm		in
Top	0.33	MIN	0.013
	0.53	MAX	0.021



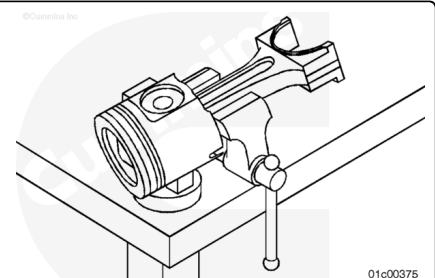
Second (full blitz chrome rings) Engine assembled before engine serial number 79383594.	0.97	MIN	0.038
	1.23	MAX	0.048
Second (semi inlaid rings) Engine assembled after engine serial number 79383594.	1.87	MIN	0.074
	2.13	MAX	0.084
Oil	0.21	MIN	0.008
	0.51	MAX	0.020

Install

Articulated Piston

⚠ CAUTION ⚠

Use a copper or aluminum vise protector when installing the connecting rod in a vise to protect the connecting rod from damage.

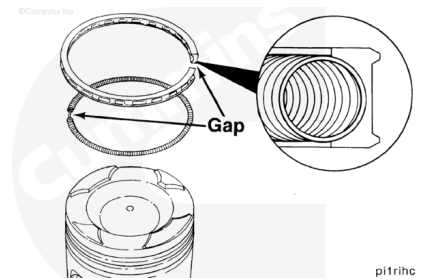


The ring gap of each ring **must not** be aligned with the piston pin or with any other ring. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.

Place the connecting rod in a vise to hold the assembly in a horizontal position while installing the rings.

A cross-sectioned view of an oil control ring is shown.

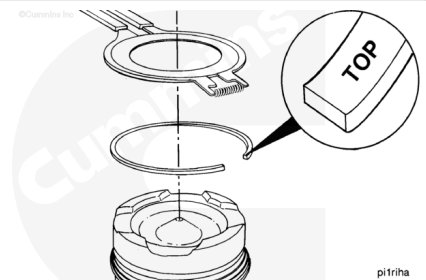
The two-piece oil control ring **must** be installed with the expander ring gap 180 degrees from the gap of the oil ring. Do **not** overlap the ends of the expander ring.



Install the piston rings with TOP toward the top of the piston.

The rings have the top marked for correct installation.

The oil control ring is symmetrical and can be installed in either orientation.

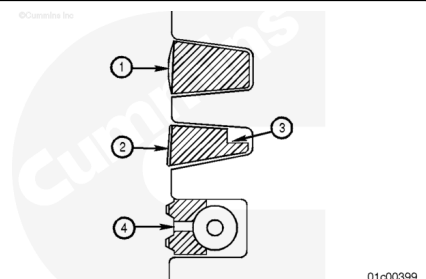


The piston **must** be held in a horizontal position to install the rings in order to keep the rings in the groove while installing the ring compressor.

For articulated pistons, the top piston ring (1) is a keystone ring with no twist or cutback notch.

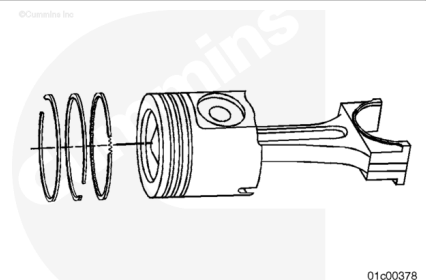
The intermediate ring (2) is a positive twist design with a cutback notch (3) on the topside and a two-degree taper face.

The oil control ring (4) is the bottom piston ring.



Rotate the rings to position the ring gaps as shown.

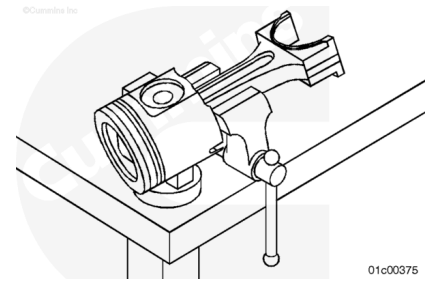
The ring gap of each ring **must not** be aligned with the piston pin, or with any other ring gap. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.



Single-Piece Piston

⚠ CAUTION ⚠

Use a copper or aluminum vise protector when installing the connecting rod in a vise to protect the connecting rod from damage.

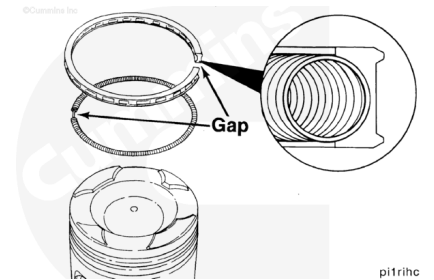


The ring gap of each ring **must not** be aligned with the piston pin or with any other ring. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.

Install the connecting rod in a vise to hold the assembly in a horizontal position while installing the rings.

A cross-sectioned view of an oil control ring is shown.

The two-piece oil control ring **must** be installed with the expander ring gap 180 degrees from the gap of the oil ring. Do **not** overlap the ends of the expander ring.

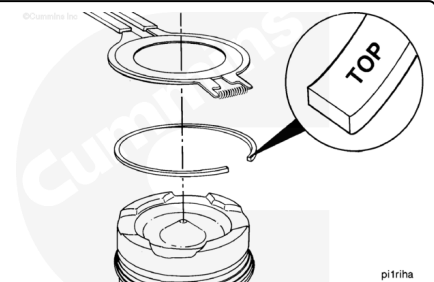


Install the piston rings with TOP toward the top of the piston.

The rings have the top marked for correct installation.

The oil control ring is symmetrical and can be installed in either orientation.

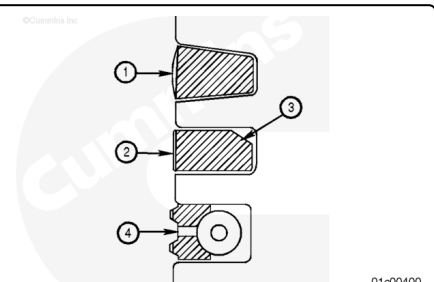
For single-piece pistons, use a universal piston ring expander to install the second ring into the piston.



The piston **must** be held in a horizontal position to install the rings in order to keep the rings in the groove while installing the ring compressor.

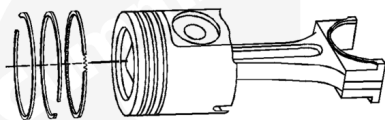
The single-piece ring set is similar to the articulated ring set, with the exception of the second ring. The second ring has a rectangular cross-section (2) with an angled cut (3) on the upper side of the inner diameter to induce a positive twist. This design improves the slobber-resistance of the ring pack.

The oil control ring (4) is the bottom piston ring.



Rotate the rings to position the ring gaps as shown.

The ring gap of each ring **must not** be aligned with the piston pin, or with any other ring gap. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.



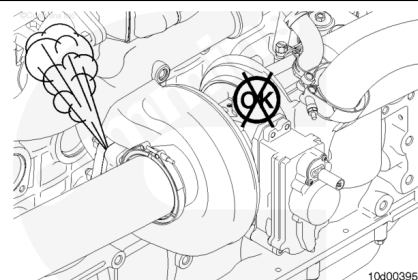
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Note : All engines must be run in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Reference Section 14 - Engine Testing for appropriate run in procedure.

Note : Use the following procedure for general run-in test overview. Refer to Procedure 014-999 in section F. (</qs3/pubsys2/xml/en/procedures/10/10-014-999.html>)

Note : When installing new piston rings with used liners, higher than average blowby may be experienced for the first several hours of operation. If excessive blowby is still seen after several hours of engine operation, consult the troubleshooting symptom tree Crankcase Gases (Blowby) Excessive in Section TS.

If malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14 (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>).



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